$\mathbf{v}_{out}^{l,n}$ (Line 3), using Logit Lens (nostalgebraist, 2020). To match the logit of each result token to the prompts that result in it (e.g., match the logit of "151" with the prompts "$1 + 150 =$","$2 + 149 =$", etc.), we convert the logits to a 2D pattern $\mathbf{L}$, where $\mathbf{L}_{ij}$ is the logit of applying the operator on i and j (Line 4). Multiplying the 2D activation pattern and 2D logit pattern element-wise (Line 5) yields the neuron's effective logit contribution to the correct answer of each prompt, i.e., the value at each index i, j marks how much the neuron promotes the answer token to the result of applying the operator on i and j. We then create two lists of prompts. One list contains prompts associated with the heuristic (Line 7), to be used as the "ground truth". We mark the number of associated prompts as k (Line 8), and find a second list, containing the top-k prompts that are most contributed to by the neuron (Line 9). We measure if the intersection of these two lists, normalized by k, is larger than the threshold $t = 0.6$ (Line 10–Line 12) to decide if the neuron implements the heuristic.

G ADDITIONAL IMPLEMENTATION DETAILS

All experimental procedures were executed using the TransformerLens library Nanda & Bloom (2022). Experiments involving Llama3-8B, GPT-J, and Pythia-6.9B were conducted on a single Nvidia-L40 GPU with 48GB of GPU memory. Llama3-70B was loaded in 16-bit precision across four Nvidia A100 GPUs, each with 80GB of GPU memory.

Arithmetic prompts. Unless otherwise specified, our experiments utilize prompts of the form $op_1 \circ op_2$=", where $op_1, op_2 \in [0, 300]$ and $\circ \in \{+, -, \times, \div\}$. The division operator ($\div$) is interpreted as integer division, considering only the integer part of the result (e.g., the ground truth result for $45 \div 4 =$" is 11). For each prompt, we compute the ground-truth result and exclude prompts whose results are not tokenized to a single token. This filtering process eliminates prompts with negative results and prompts for which the correct answer exceeds the single-token limit (The highest token from which the model splits numbers into two tokens). The single-token limit s for Llama3-8B and Llama3-70B is $s_{llama3} = 1,000$, while for GPT-J and Pythia-6.9B, it is $s_{gptj} = s_{pythia} = 520$.

Circuit discovery via activation patching. In Section 2.1, we quantify the effect of patching on probabilities. We find that applying our effect measure (Equation (1)) on logits instead of probabilities, does not alter the result significantly—the most effective components remain consistent, differing only in the scale of the effect.

Circuit faithfulness evaluation. Thus far, to measure the faithfulness of the circuit, we have performed mean ablation on all non-circuit components. For that, we calculate the mean activation output of each component, across all arithmetic prompts. In this context, we do not filter correct prompts exclusively, but instead use *all* prompts of the form "$op_1 \circ op_2$=", for $op_1, op_2 \in [0, 300]$. This leads to an equal amount of prompts per operator, thus maintaining the mean activation balanced across the different operators. When evaluating a circuit with partial MLP layers (Section 3.1), we mean ablate lower-effect neurons in middle- and late-layer MLPs only, starting and ending in the earliest and latest layers from which the answer is extractable via linear probe (Section 2.2). These layer ranges are $[16, 32], [39, 80], [14, 32], [17, 28]$ In Llama3-8B, Llama3-70B, Pythia-6.9B, and GPT-J, respectively.

Linear probing for correct answers. We define a linear probe $f_{l,p} : \mathbb{R} \to \mathbb{R}^s$, where s is the single-token limit, to predict the correct answer token from the output representation at layer l and position p. We compute these outputs by using all correctly completed prompts, with 80% used for training the classifier and 20% for evaluation. Each probing classifier is implemented as a one-layer fully connected model. We train it using the Adam optimizer (Kingma & Ba, 2015) with a learning rate of 0.0003 and a batch size of 32, optimizing a cross-entropy loss function.

Heuristic classification. In applying the heuristic classification algorithm (Section 4.1), we use all activations from the last position $p = 4$ and employ a classification score threshold $t = 0.6$. Due to a lack of ground truth data for neuron-to-heuristic matches, the threshold is chosen to achieve a Pareto-optimal balance of false positives, that occur more for a lower threshold, and false negatives, that occur more in a higher threshold. The amount of false classifications is estimated manually.

Table 3: Accuracy of the analyzed models on arithmetic prompts.

	Operator				
Model	$+$	$-$	$\times$	$\div$	**Average**
Llama3-8B	0.97	0.96	0.84	0.92	0.95
Llama3-70B	0.97	0.99	0.99	0.73	0.88
Pythia-6.9B	0.30	0.04	0.27	0.75	0.43
GPT-J	0.23	0.09	0.46	0.64	0.37

H MODEL ACCURACIES ON ARITHMETIC PROMPTS

Table 3 presents the accuracy of the analyzed models on arithmetic prompts. The accuracy is evaluated across prompts where the result is represented by a single token, with both operands constrained to the interval $[0, 300]$, where 300 represents the maximum value of operands in the analyzed prompts (Section 2.1), chosen for efficiency. It is noteworthy that the high accuracy rates observed in division operations for the smaller-scale models (GPT-J, Pythia-6.9B) can be attributed to the non-uniform distribution of answers in integer division, i.e. - half of the legal prompts result in the token '0'.

I RESULTS ON ADDITIONAL MODELS

To demonstrate the generalizability of our findings across differently-trained LLMs, we conduct our primary experiments on Llama3-70B (Dubey et al., 2024), Pythia-6.9B (Biderman et al., 2023), and GPT-J (Wang & Komatsuzaki, 2021). We replicate the experiments for circuit discovery and evaluation (Section 2.1), linear probing for answer token embeddings (Section 2.2), top-k neuron faithfulness analysis (Section 3.1), and heuristic analysis (Section 4.2). The results obtained are similar to those of Llama3-8B across all three LLMs:

- The circuit comprises a sparse subset of attention heads that project operand and operator information to the final position, along with all MLP layers. Early-layer MLPs process information at the operand and operator positions, while middle- and late-layer MLPs process the combined information at the last token. These findings are illustrated for Llama3-70B (Figure 16a), Pythia-6.9B (Figure 17a), and GPT-J (Figure 18a).
- Linear probing results indicate that the correct answer can only be extracted with high accuracy in the final position, following the processing initiated by middle-layer MLPs. These observations are presented for Llama3-70B (Figure 16b), Pythia-6.9B (Figure 17b), and GPT-J (Figure 18b).
- The circuit requires a sparse subset of middle- and late-layer MLP neurons to achieve maximal faithfulness. These results are depicted for Llama3-70B (Figure 16c), Pythia-6.9B (Figure 17c), and GPT-J (Figure 18c).
- The heuristic neurons in the middle- and late-layer MLPs are the model components that write the correct answer to the model output. This is shown by repeating the prompt-guided knockout experiment (Section 4.2). The ablation of specific heuristic neurons associated with prompts results in a more significant reduction in model accuracy compared to the ablation of a random set of neurons of equivalent size. These findings are illustrated for Llama3-70B (Figure 8), Pythia-6.9B (Figure 17d), and GPT-J (Figure 18d).

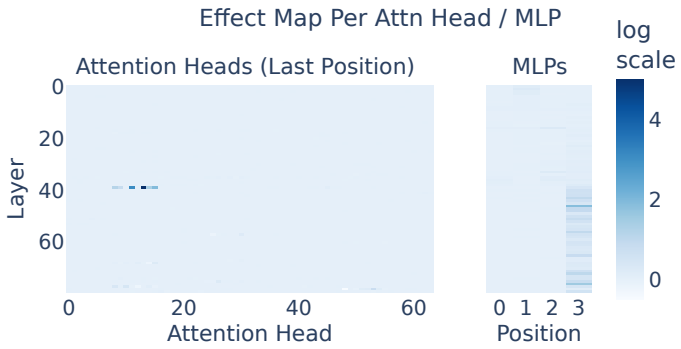
The results observed in Llama3-70B exhibit the highest similarity to those reported in Llama3-8B, with a more pronounced knockout effect compared to the other two models. We hypothesize that this indicates a more sophisticated development of the bag of heuristics in Llama3-70B, potentially facilitated by its larger size and modern training methodology.

J ADDITIONAL EXAMPLES FOR ARITHMETIC HEURISTICS

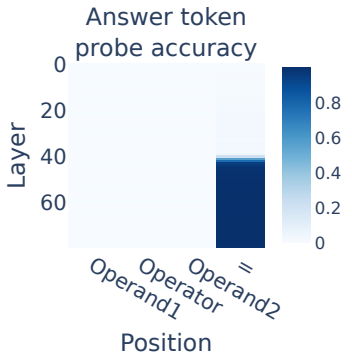
We present supplementary examples of heuristic neurons (Table 4, Table 5, Table 6, and Table 7 for the four arithmetic operators, respectively). Each neuron $\mathbf{h}^{l,n}_{post}$ found to be important for a specific

arithmetic operator (Section 3.1) is categorized as one or more heuristics. For each designated operator, we randomly present four neurons. We compute and report each neuron's activation pattern, the ten numerical tokens with the strongest embeddings in each neuron's value vector $\mathbf{v}_{out}^{l,n}$, and provide a non-exhaustive list of the heuristics it implements.

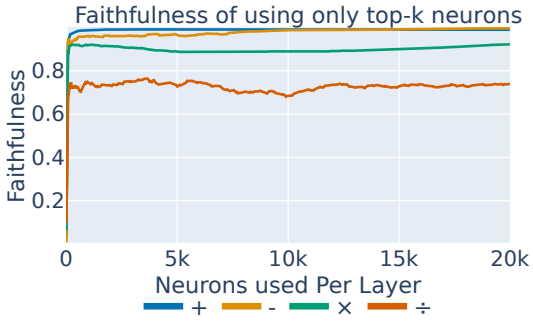
Furthermore, we present in Table 8 several examples of causally significant neurons that have not been classified as specific heuristics. These neurons are relevant to the aforementioned limitation in our methodology (Section 8); it is conceivable that they too may be considered components of the bag of heuristics, depending upon our ability to comprehend the abstractions they implement.



(a) Llama3-70B localization results.



(b) Llama3-70B linear probing results.



(c) Llama3-70B faithfulness as function of MLP neurons in each layer.

Figure 16: Llama3-70B analysis results